

DA6360 : Worksheet 5

- (1) Let $x \in \mathbb{R}^3$ be a some given vector. It is known that for this vector, $\|x\|_\infty \leq 5$. Find the smallest constant C such that $\|x\|_1 \leq C$
- (2) Draw the set $B_\infty(x_0, 3) \cap B_2(x_1, 3)$, where $x_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $x_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$
- (3) Consider a point $x = [11]$ in the standard basis. Let x_V be its representation in the basis $V = \{v_1 = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, v_2 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}\}$. Find x_V
- (4) Let T be a linear transformation from $\mathbb{R}^3 \rightarrow \mathbb{R}^3$. We know that $T(e_1) = e_2, T(e_2) = e_3$ and $T(e_3) = e_1$. Let M_T be the matrix representation of T in the standard basis. Find M_T and M_T^2 .
- (5) A real symmetric $H \in \mathbb{R}^{d \times d}$ can be decomposed as $H = VDV^\top$ where D is a diagonal matrix whose i^{th} diagonal entry is the i^{th} eigenvalue of H and $V = [v_1 \dots v_d]$ is the matrix whose i^{th} is the corresponding eigenvector. For the matrix $H = \begin{bmatrix} 0.5 & 1.5 \\ 1.5 & 0.5 \end{bmatrix}$ find V and D . Is the matrix H positive definite?
- (6) Given 5 examples of 3×3 positive definite real symmetric matrices.
- (7) Let $f_1(x) = \frac{1}{2}(5x(1)^2 + x(2)^2)$, and $f_2(x) = \frac{1}{2}(5x(1)^2 - x(2)^2)$.
 - (a) Give first and second order approximation for f_1 and f_2 at $\hat{x} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$.
 - (b) Give the set of directions at $\hat{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ that will decrease f_1 .
 - (c) Give the set of directions at $\hat{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ that will increase f_2 .
 - (d) Draw the contours of f_1 and f_2 .
- (8) For $f(x) = x(1) \exp(-(x(1)^2 + x(2)^2))$
 - (a) Plot the function in a Jupyter notebook to get an idea of how it looks like (plotting question will not be examination).
 - (b) Find the gradient and stationary points (i.e, x_* such that $\nabla f(x_*) = 0$).
 - (c) Find the first and second order approximation of f at $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -1 \end{bmatrix}$.
 - (d) Find the local minima and local maxima.
 - (e) Draw the contours of f .
 - (f) Give the set of directions at $\hat{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ that will decrease f .
 - (g) Give the set of directions at $\hat{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ that will increase f .
- (9) $f(x) = (x(2) - x(1)^2)^2 + x(1)^5$
 - (a) Plot the function in a Jupyter notebook to get an idea of how it looks like (plotting question will not be examination).
 - (b) Find the gradient and stationary points (i.e, x_* such that $\nabla f(x_*) = 0$).
 - (c) Find the first and second order approximation of f at $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} -1 \\ -1 \end{bmatrix}$.
 - (d) Find the local minima and local maxima.
 - (e) Give the set of directions at $\hat{x} = \begin{bmatrix} 2 \\ 2 \end{bmatrix}$ that will decrease f .

- (f) Give the set of directions at $\hat{x} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$ that will increase f .